SCG-HMH: further physics exploitations the design opens up

Supplement to the SCG-HMH Closed Derivative Chain, v8 (V10 release package)

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New in V9, derived here, not from prior records. This supplement records ten optional, downstream or site-level uses of what an SCG-HMH module already produces, ranked most meaningful first, each with its status (new derivation / implied by design / speculative). It uses no seeded plasma, no combustion and no hot ionisation, adds no parasitic load inside the 0.8 kW module budget, and changes no booked number. Items that would touch the module itself are flagged "would need Morgan's approval". Nothing here has been demonstrated on an SCG-HMH bench.

Ground rules (Morgan, 16:10–16:11): "don't kill the design", "no seeded plasma"

Every item below works with the published v8 design as it stands: - no seed (alkali, caesium, potassium), no combustion and no hot thermal ionisation; - no change to the working fluid, temperatures, operating point or any booked number (99.55 MW per module at 5 T, 150,938 rpm, SF 2; COP_th 0.2175; COP_e 124,437; train 9.9549 GW); - no new parasitic or hardware load inside the module's 0.8 kW budget.

Each item uses only outputs the module already produces: its DC power, its LN_2 and cold, the waste-heat reset streams, the 0.4 % vent and makeup stream, and the rotor's stored energy. They are **optional downstream or site-level uses**. Anything that would touch the module itself (a probe, a port or a change to a stream's route) is flagged **"would need Morgan's approval"**. Ideas that would need the core changed were dropped (see the end).

Already covered elsewhere, so skipped: transistor-style gain, the CV^2f chip-heat loop, cryo-CMOS and the Landauer floor (Transistor supplement); district, greenhouse and desalination-preheat heat (v8 Table, 6.6.10); food, water, DAC and space at COP_th 0.8266 (v8 §12.2); grid flywheels as a rotor benchmark (companion v4/v5).

Labels. *Derived here* means computed in this note from v8 numbers and public constants; CoolProp 7 was used for the N_2 and H_2 properties. Status tags: **implied by design** (the records already contain the feature and the use follows

directly), **new derivation** (new arithmetic on v8 numbers plus a public benchmark), **speculative** (physically sound, but it depends on unknowns or would need approval).

A fact derived here that underpins items 1–3: the ring is a low-voltage, mega-amp DC source

The ring is a Faraday-type (homopolar) MHD generator, so its output is DC. At the booked point, v = ωr = 15,806.19 rad/s × 0.093 m = 1,469.98 m/s: - open-circuit gap voltage vBδ = 1,469.98 × 5 × 0.002 = **14.70 V** (v8 Table 13y gives the same formula as 19.48 V at 200,000 rpm); - terminal voltage K·vBδ = 0.65 × 14.70 = **9.55 V**; - current I = P / V = 99.55 MW / 9.55 V = **10.42 MA** per module.

[derived here] v8 does not size the output bus. This changes no booked number. It does say what kind of loads suit the module best: loads that want very high DC current at a few volts.

Ranked list (most meaningful first)

1. Direct DC electrochemistry: green hydrogen, aluminium, chlor-alkali, electrowinning

1. **Principle.** Electrolysis runs on DC at a few volts per cell. PEM water cells run at about 1.8–2.0 V. Hall–Héroult aluminium cells run at 4.15–4.64 V and 300–600 kA, in potlines of 100–400 cells in series. Today grid AC is transformed and rectified to reach this.
2. **Design feature.** The homopolar ring's native output of 9.55 V and 10.42 MA DC.
3. **Estimate [derived here].** About 9.55 V spans about two aluminium cells or about five PEM cells in series. The 10.42 MA is about 17–35 potline currents of 300–600 kA in parallel. Rectifier and transformer stages could in principle be skipped. One module's 99.55 MW at 50–55 kWh/kg H_2 (IRENA system range starts at 50; up to 83) gives **43.4–47.8 t H_2 per day**. The thermodynamic floor of 39.4 kWh/kg (HHV) would give 60.6 t/day.
4. **Benchmarks.** Carmo et al. 2013, PEM electrolysis review, https://doi.org/10.1016/j.ijhydene.2013.01.151. IRENA 2020, *Green hydrogen cost reduction* (system consumption 50–83 kWh/kg for PEM), https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2020/Dec/IRENA_Green_hydrogen_cost_2020.pdf. *The Aluminum Smelting Process and Innovative Alternative Technologies* (PMC open access; potlines 300–600 kA, 100–400 cells in series), https://pmc.ncbi.nlm.nih.gov/articles/PMC4131935/. Obaidat et al. 2018 (typical cell voltage 4.64 V), https://doi.org/10.3390/su10041216.
5. **Status: new derivation.** Site-level; the module is untouched. Open engineering point: how 10 MA is collected from the strip electrodes (see item 2).

2. HTS DC busbar on the same LN_2 (the output's natural partner)

1. **Principle.** REBCO carries DC with zero resistance at 77 K. Copper at 10 MA is costly: at 1 A/mm², 10.42 MA needs about 10.4 m² of copper and dissipates ρJI ≈ **175 kW per metre**, about 0.18 % of the module's output per metre [derived here; ρ_Cu = 1.68×10^{-8} Ω·m].
2. **Design feature.** The module already holds REBCO technology and a 77.35 K LN_2 tank. Any heat leak into an HTS bus boils LN_2, which the cycle needs boiled anyway (vaporiser duty). That is a qualitative synergy and is not booked.
3. **Estimate [derived here].** 10.42 MA ÷ 190 kA (DEMO200's demonstrated current) ≈ **55 DEMO200-class busbars** per module, or about 52 at the 200 kA rating.
4. **Benchmarks.** DEMO200 (VESC, TRIMET, Messer, KIT): a REBCO busbar in boiling LN_2 at 77 K carried 170 kA for hours and peaked at 190 kA at the Voerde aluminium smelter in September 2024. Its precursor "3S" carried 20 kA in BASF chlorine electrolysis. Sources: https://www.vesc-superbar.de/wp-content/uploads/2024/11/PressRelease_Demo200_241010_en.pdf and https://demo200.de/2024/10/we-did-it/; design paper https://doi.org/10.1109/TASC.2022.3152987 (its abstract names aluminium smelters, electrolysis plants *and large data centres* as targets). TRIMET Hamburg SuprAl: a 600 m, 200 kA HTS bus with >90 % lower transmission loss, due mid-2027, https://www.trimet.eu/en/press/whats-new/news-detail/after-world-record-superconductor-ready-for-industrial-use. AmpaCity Essen: 1 km, 10 kV, 40 MW HTS cable in grid service from 2014 for seven years, https://www.kit.edu/kit/english/pi_2014_15058.php. Copper resistivity: Matula 1979, https://doi.org/10.1063/1.555614.
5. **Status: new derivation.** Site-level. The bus is downstream of the terminals, so nothing is added inside the 0.8 kW budget. **Worth Morgan's attention:** the v8 package does not yet say how the 10 MA leaves the ring. This is a sizing question, not a change to the design.

3. Green ammonia using the nitrogen already on site

1. **Principle.** Haber–Bosch: $N_2 + 3H_2 \rightarrow 2NH_3$. Conventional plants pay for an air-separation unit to make N_2, and for natural-gas reforming to make H_2. The IEA puts ammonia at 185 Mt in 2020, about 2 % of final energy use and 450 Mt CO_2 of direct emissions.
2. **Design feature.** The module already buys, liquefies, stores and vents pure N_2: makeup $\dot{m}_{mu}$ = 0.4 % × 410.52 kg/s = **1.642 kg/s**, and the vent leaves before the J–T HX at 453 K. The H_2 comes from item 1, on the module's own DC.
3. **Estimate [derived here].** Turning one module's whole output into H_2 (50–55 kWh/kg) gives ammonia that needs 2.33–2.56 kg/s of N_2. The module's own makeup and vent stream of 1.642 kg/s is **64–71 %** of that. At full output this is **244–269 t NH_3 per day per module**. The 1.642 kg/s alone, stoichiometrically, makes 172 t NH_3 per day using 0.354 kg/s H_2, which takes 64 MW at 50 kWh/kg (64 % of one module). A train of 100 modules at the booked output would be about 24,400–26,900 t/day, about 8.9–9.8 Mt/yr if run continuously, or about 5 % of 2020 world output. Haber–Bosch loop

compression is not included. Electrolysis also releases 8 kg of O_2 per kg of H_2, about **350–380 t O_2 per day per module**, as a co-product.
4. **Benchmarks.** IEA *Ammonia Technology Roadmap* (2021), https://www.iea.org/reports/ammonia-technology-roadmap (DOI 10.1787/f6daa4a0-en). This links straight to v8 §12.2's "fertiliser synthesis" line, giving it a mechanism and a number.
5. **Status: new derivation.** The N_2 and the H_2 path are site-level. Capturing the vent stream instead of releasing it to air **would need Morgan's approval**, because it changes where the vent line ends. It adds no load, and the 0.373 MW vent heat stays booked. The purity of the vent N_2 after the plasma pass is unmeasured, so this is a bench item.

4. The rotor as a flywheel: ride-through and fast frequency response

1. **Principle.** Stored kinetic energy $E = \frac{1}{2}I\omega^2$. The grid measures this as the inertia constant H = E / P_rated. Typical plant runs at about 2–7 s (NREL).
2. **Design feature.** A shaftless rotor levitated on YBCO bearings at 150,938 rpm, with no bearing friction to waste the store (L1 drag ≈ 9.7 W).
3. **Estimate [derived here].** This uses the v8 r22c rotor, which is assumed and not published: 47.69 kg, a titanium core of radius 0.0910 m (46.89 kg) and a 0.80 kg CFRP sleeve at 0.098–0.100 m, 0.4 m long. I = 0.2020 kg·m². At 15,806 rad/s, **E = 25.2 MJ (7.0 kWh) per module**, **H ≈ 0.25 s** at 99.55 MW. At 13 T and 197,761 rpm, E is 43.3 MJ (H ≈ 0.04 s at 1,155 MW). A train of 100 holds about 2.5 GJ. Dropping from 100 % to 90 % speed releases 4.8 MJ. H is about ten times below a synchronous plant's, so the honest value is **fast ride-through** (around 0.25 s of full output to bridge faults and switching), not bulk storage. Two notes. The output is DC, so the grid sees this store only through a grid-forming inverter. And because P rises faster than rpm², drawing on the store lowers output steeply, so it acts as a self-limiting buffer.
4. **Benchmarks.** NREL, *Inertia and the Power Grid: A Guide Without the Spin* (2020), https://www.osti.gov/biblio/1659820 (DOI 10.2172/1659820). Companion v4/v5 already cite the Beacon Stephentown flywheel fleet (20 MW) and the Boeing HTS-bearing flywheel.
5. **Status: new derivation.** It fits the records' "Edge-AI safety flywheel" and the "grid-flywheel multiplier" amplifiers (v8 Table 14, #5 and #61), which name the idea but give no figure. It is a control choice in the AI-oracle; the rotor mass is assumed.

5. The shared LN_2 tank as a liquid-nitrogen energy store (LAES-type)

1. **Principle.** LN_2 is stored exergy. The minimum work to liquefy N_2 from 300 K and 1 atm is **769.1 kJ/kg (0.2136 kWh/kg)** [derived here, CoolProp]. Liquid-air storage charges a cryogenic tank and discharges it by boiling and expanding the liquid with low-grade heat.
2. **Design feature.** The design already has a shared 77.35 K tank, a waste-heat-led LN_2 generator and a waste-heat expansion channel. That makes it, structurally, a waste-heat-assisted LAES plant.
3. **Estimate [derived here].** The exergy ceiling is about **214 MWh per 1,000 t of LN_2** stored. Practical recovery would be lower. The pilot plant below

reports about 60 % round-trip efficiency as the projected optimum.
4. **Benchmarks.** Morgan, Nelmes, Gibson & Brett 2015, *Liquid air energy storage – analysis and first results from a pilot scale demonstration plant*, Applied Energy 137, 845, https://doi.org/10.1016/j.apenergy.2014.07.109. It reports that **45 % of supplied low-grade heat was converted to work** in discharge, and that the pilot ran on tanker LN_2. Highview Power Carrington, 50 MW / 300 MWh, under construction, https://highviewpower.com/projects/.
5. **Status: implied by design / speculative.** Storing surplus LN_2 beyond the loop inventory means an oversized site tank. That is site-level and **would need Morgan's approval**, because the tank is shared with the module. No booked number changes.

6. The cold end as an industrial cold sink: freeze desalination, LH_2 pre-cooling, cryogenic processing

1. **Principle.** The cycle has to put heat *into* its 77 K liquid to boil it. Any site process that needs heat *removed* can in principle supply part of that, as LNG terminals do in "cold-energy utilisation".
2. **Design feature.** LN_2 boiled at 77.35 K (h_fg 199.18 kJ/kg), plus sensible warming. 77.35 K → 290 K takes **422.8 kJ/kg** [derived here].
3. **Estimate [derived here, ideal heat balance only].**
   - Freeze desalination: up to 422.8 / 333.6 ≈ **1.27 kg of ice per kg of LN_2** warmed.
   - H_2 liquefaction pre-cool (normal H_2 at 20 bar, 300 → 80 K: 2,924 kJ/kg): about **6.9 kg of LN_2 per kg of H_2**, the same LN_2 pre-cool stage used in real plants. This pairs with items 1 and 3 for LH_2 export.
   - Cryogenic grinding and recycling, and cold-chain freezing: qualitative.
4. **Benchmarks.** *Freeze desalination of seawater using LNG cold energy*, Water Research, https://doi.org/10.1016/j.watres.2016.06.046 (about 300 ppm TDS product ice). Linde Ingolstadt and Leuna LH_2 plants use LN_2 pre-cooling to about 80 K (Ingolstadt 13.58 kWh/kg, Leuna 11.9 kWh/kg): *Hydrogen liquefaction: a review of the fundamental physics, engineering practice and future opportunities*, Energy Environ. Sci. 15, 2690 (2022), https://doi.org/10.1039/D2EE00099G.
5. **Status: speculative.** It would substitute a low-temperature heat source for part of the booked waste heat at the cold end. That could move the cycle's heat split, so it **would need Morgan's approval** and a re-run of the 6.6 books before anything is booked. The cleanest form uses only *surplus* LN_2 (item 5), which leaves the module untouched.

7. Air-separation co-products from the makeup N_2 supply

1. **Principle.** Separating air for N_2 also produces O_2 and Ar. Standard dry-air mass fractions: N_2 75.52 %, O_2 23.14 %, Ar 1.29 %.
2. **Design feature.** The records' PSA/membrane makeup supply (v8 Table 14, L7, 0.18 kWh/kg).
3. **Estimate [derived here].** At 1.642 kg/s of N_2 makeup, full recovery would give about **43 t O_2 per day and 2.4 t Ar per day per module**. PSA and membranes give an O_2-enriched reject, not pure O_2. Pure co-products need a cryogenic air-separation unit, which the site's LN_2 cold would favour.
4. **Benchmark.** Standard air composition; no project citation.

5. **Status: speculative.** Site-level and small. It ranks below item 3, where electrolysis gives about 8× more O_2.

8. Centrifugal isotope enrichment (^{15}N) in the cold, fast-spinning gas

1. **Principle.** A gas in solid-body rotation stratifies by mass. The equilibrium radial separation factor is $\alpha = \exp(\Delta m \cdot v^2/(2kT))$, or across a thin gap $\alpha = \exp(\Delta m \cdot \omega^2 \cdot r \cdot \delta/(kT))$. Cold and fast is the best combination.
2. **Design feature.** Cold N_2 (about 80 K) at a 1,470 m/s tip speed.
3. **Estimate [derived here; Δm = 1 u for $^{14}N^{15}N$ against $^{14}N_2$, T = 80 K].**
   - Across the 2 mm gap at r = 0.093 m: **$\alpha \approx 1.072$**.
   - Axis-to-rim equivalent: $\alpha \approx 5.07$, but the gas sits only in the gap, so this is not available.
   - Reference: a UF_6 centrifuge at 600–700 m/s and 300 K, with Δm = 3 u, gives 1.24–1.34 on the same formula.
   - Natural ^{15}N is 0.366 %. It is used as a tracer and proposed for nitride nuclear fuels.
4. **Benchmarks.** Olander 1981, gas-centrifuge theory, Prog. Nucl. Energy 8, 1, https://doi.org/10.1016/0149-1970(81)90026-3. Krishnan, Geva & Hirshfield 1981, *Plasma Centrifuge*, PRL 46, 36, https://doi.org/10.1103/PhysRevLett.46.36 (rotation up to 7.4 km/s, enrichment up to ×2 for ^{65}Cu).
5. **Status: speculative.** It assumes the gas co-rotates with the plasma, which is not shown. Drawing inner and outer gap samples would add ports to the channel, so it **would need Morgan's approval**. At most this is a bench diagnostic or a niche co-product.

9. The stators as a field-energy buffer (SMES-type)

1. **Principle.** Magnetic energy density $B^2/(2\mu_0)$.
2. **Design feature.** The four ReBCO stators at 5 T (13 T upgrade, 20 T ceiling) and their self-excited field.
3. **Estimate [derived here].** 9.95 / 67.2 / 159.2 MJ/m³ at 5 / 13 / 20 T. The stator bore volume is not published, so as a scale only: the 0.2 m × 0.4 m rotor envelope (0.0126 m³) would hold 0.125 / 0.845 / 2.0 MJ. That is far below the rotor's 25 MJ. There is also a basic conflict: drawing energy from the field lowers B and output, because $P \propto B^2$. The useful role is limited to quench and fault energy management, which v8's r23 safety note already covers.
4. **Benchmarks.** *A study of the status and future of superconducting magnetic energy storage in power systems*, Supercond. Sci. Technol. 19 (2006) R31, https://doi.org/10.1088/0953-2048/19/6/R01. Tixador 2008, *Superconducting Magnetic Energy Storage: Status and Perspective*, https://hal.science/hal-00364089.
5. **Status: speculative, weak.** Kept only so it is on record that it was checked. Item 4 is the better store.

10. A rotating, magnetised, collisional plasma flow as a laboratory (astrophysical-analogue and plasma-physics data)

1. **Principle.** Lab Couette flows probe accretion-disk angular-momentum transport, including the magnetorotational instability (MRI).
2. **Design feature.** A high-field (5–20 T), high-speed, dense cold plasma ring.
3. **Estimate [derived here].** Magnetic Reynolds number $Rm = \mu_0 \sigma v L$, with σ = 17,328 S/m and v = 1,470 m/s:
   - **0.064** on the 2 mm gap scale: the flow barely distorts the stator field, which is good for the design;
   - **≈ 3.0** on the 93 mm radius scale. That is below the ≈ 25 reached in the Plasma Couette Experiment and the critical-Rm regime used for the MRI observation, so SCG-HMH is **not** an MRI analogue at the booked point. What it would offer is data in an unusual corner (very high B, high pressure, cold heavy particles), for example on viscosity and the Hall regime.
4. **Benchmarks.** Wang et al. 2022, *Observation of Axisymmetric Standard MRI in the Laboratory*, PRL 129, 115001, https://doi.org/10.1103/PhysRevLett.129.115001 (arXiv https://arxiv.org/abs/2209.08457). Collins et al. 2012, *Stirring Unmagnetized Plasma*, PRL 108, 115001, https://doi.org/10.1103/PhysRevLett.108.115001 (arXiv https://arxiv.org/abs/1205.6704; Rm ≈ 25).
5. **Status: speculative.** It needs only the bench's existing passive diagnostics (V, I, Hall probe). Any optical port or added probe **would need Morgan's approval**.

Checked and dropped (they would break the rules or are weak)

- **Plasma nitrogen fixation (NOx) and ozone.** These need O_2 in the plasma, and v8 removes O_2 on purpose (amplifier #86, anti-attachment). Diverting the plasma or changing the gas would change the core. Dropped; ammonia via item 3 gives the fertiliser route without touching the plasma.
- **Any seeded or hot-plasma use.** Excluded by rule.
- **MRI-class field access for materials processing.** The bore is occupied by the rotor and the plasma, so there is no access without changing the module. Dropped.
- **MHD thrusters and propulsion.** That is just using the electricity; the ship-motor benchmark is already in the companion. Dropped as not new physics.
- **Hosting quantum or HTS-SQUID sensors at 77 K.** The strong stray field near the stators makes this weak, and cryo-electronics are covered by the Transistor supplement. Dropped.
- **Regenerative-braking analogue.** Folded into item 4 (spin-down returns rotor energy as electricity).
- **Space.** Already in v8 (COP_th 0.8266, TITAN). Not repeated.

What this changes in V9, and the open question

Nothing is booked. Every v8 number is unchanged. The one point to put to Morgan is a question, not a change: the 10.42 MA / 9.55 V DC output (derived here) has no output-bus sizing in the package. That is where HTS busbars and direct electrolysis fit.

Open question (added to the record): the V9 package never sizes the output bus, i.e. how the 10.42 MA at 9.55 V DC per module is collected from the strip electrodes and carried off the ring. This is a sizing question for the bench and site design, not a change to the design.

Link check (29 October 2026, about 16:15 London time)

- **Returned 200 by curl this session:** the IRENA PDF; doi 10.1016/j.ijhydene.2013.01.151; the PMC Kvande article; the IEA roadmap; the VESC press release; demo200.de; the TRIMET release; KIT's AmpaCity release; doi 10.1016/j.apenergy.2014.07.109; the Highview projects page; OSTI 1659820 (NREL, title confirmed); doi 10.1016/j.watres.2016.06.046; doi 10.1016/0149-1970(81)90026-3; doi 10.1088/0953-2048/19/6/R01; arXiv 2209.08457 and 1205.6704.
- **Returned 202:** doi 10.1109/TASC.2022.3152987.
- **Blocked automated checks (403):** APS (PRL 46 36, 129 115001, 108 115001), RSC (D2EE00099G), MDPI (su10041216) and AIP (10.1063/1.555614). Each was found live through a web search this session, with matching titles and metadata.
- The hal.science SMES link came from search and was not curl-checked.

All sources (consolidated)

1. https://doi.org/10.1016/j.ijhydene.2013.01.151
2. https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2020/Dec/IRENA_Green_hydrogen_cost_2020.pdf
3. https://pmc.ncbi.nlm.nih.gov/articles/PMC4131935/
4. https://doi.org/10.3390/su10041216
5. https://www.vesc-superbar.de/wp-content/uploads/2024/11/PressRelease_Demo200_241010_en.pdf
6. https://demo200.de/2024/10/we-did-it/
7. https://doi.org/10.1109/TASC.2022.3152987
8. https://www.trimet.eu/en/press/whats-new/news-detail/after-world-record-superconductor-ready-for-industrial-use
9. https://www.kit.edu/kit/english/pi_2014_15058.php
10. https://doi.org/10.1063/1.555614
11. https://www.iea.org/reports/ammonia-technology-roadmap
12. https://www.osti.gov/biblio/1659820
13. https://doi.org/10.1016/j.apenergy.2014.07.109
14. https://highviewpower.com/projects/
15. https://doi.org/10.1016/j.watres.2016.06.046
16. https://doi.org/10.1039/D2EE00099G
17. https://doi.org/10.1016/0149-1970(81
18. https://doi.org/10.1103/PhysRevLett.46.36
19. https://doi.org/10.1088/0953-2048/19/6/R01

20. https://hal.science/hal-00364089
21. https://doi.org/10.1103/PhysRevLett.129.115001
22. https://arxiv.org/abs/2209.08457
23. https://doi.org/10.1103/PhysRevLett.108.115001
24. https://arxiv.org/abs/1205.6704
25. DOI 10.1787/f6daa4a0-en (https://doi.org/10.1787/f6daa4a0-en)
26. DOI 10.2172/1659820 (https://doi.org/10.2172/1659820)